

AIScraperAgent

AI-Driven Automated Monitoring & Vector Search System for Public Procurement (AMI/AO)

Surveillance automatisée et recherche vectorielle sémantique pour les marchés publics — Phase 1



Zero-maintenance API discovery

Découverte automatique des endpoints API

01 / INNOVATION



Local Qdrant vector store

Stockage vectoriel local Qdrant

02 / INNOVATION



Real-time network interception

Interception réseau en temps réel

03 / INNOVATION

Problem & Vision

Problématique & Vision du projet

← CHALLENGE → SOLUTION

⚠ THE CHALLENGE

Manual monitoring is slow & fragile



Slow, fragmented monitoring prone to missed deadlines

Surveillance manuelle lente et dispersée



Traditional DOM scrapers break when React/Angular restructures pages

Scrapers DOM classiques cassent au moindre changement



No unified search across heterogeneous portals (ministries, regulators, World Bank)

Aucune recherche unifiée entre portails hétérogènes



💡 THE SOLUTION

Dynamic interception + AI semantic scoring



Intercept live XHR/Fetch network responses via Playwright

Interception des réponses XHR/Fetch via Playwright



AI evaluates backend APIs using sentence-transformer embeddings

Évaluation sémantique des APIs backend par embeddings



Unified schema + Qdrant vector search across all sources

Schéma unifié + recherche vectorielle Qdrant

Technology Stack

Stack technique & outillage

5 LAYERS · BILINGUAL



L1

Core Runtime

Exécution

- > Python 3.10+
- > FastAPI backend service



CORE

Dynamic Interception

Interception dynamique

- > Playwright sync_playwright (SPA)
- > BeautifulSoup + Requests (static)
- > Feedparser (RSS)



L3

AI & Embeddings

IA & Embeddings

- > sentence-transformers all-MiniLM-L6-v2
- > 384-dim dense vectors
- > scikit-learn cosine similarity



L4

Vector + Relational Storage

Stockage vectoriel + relationnel

- > Qdrant Vector DB (384-dim cosine index)
- > SQLite via SQLAlchemy ORM



L5

API & Exploitation Layer

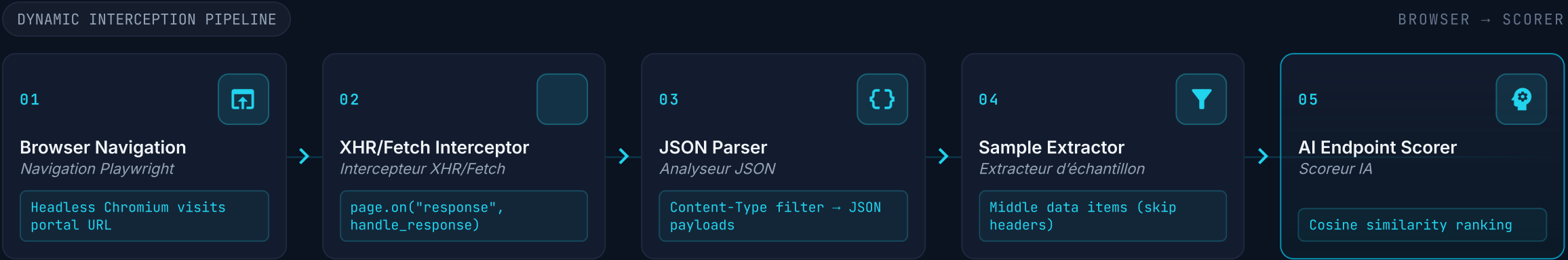
Couche API & exploitation

- > FastAPI REST endpoints
- > Semantic search service
- > JSON schema normalization pipeline

System Architecture & Interception Pipeline

Architecture & pipeline d'interception dynamique

5-STAGE FLOW



Why not DOM scraping?

React/Angular DOM shifts on every build. Network responses stay stable.

Le DOM change à chaque build ; les réponses réseau restent stables.



Safe payload sampling

Middle JSON items avoid headers/metadata noise for cleaner embeddings.

Les items centraux évitent le bruit des en-têtes et métadonnées.

AI-Driven Endpoint Discovery & Scoring

Découverte & scoring automatique des endpoints API

MECHANISM

- 01

Build domain usefulness prompt (software, AI/ML, cybersecurity, digital transformation keywords)
Prompt de pertinence domaine (logiciel, IA/ML, cybersécurité, transformation digitale)
- 02

Encode JSON sample into 384-dim sentence-transformer vector
Encodage en vecteur 384-dim via sentence-transformers
- 03

Cosine similarity vs domain query vector
Similarité cosinus vs vecteur de requête domaine
- 04

Auto-select top-ranked endpoint — no hardcoded API paths
Sélection automatique du meilleur endpoint — zéro hardcodage

RANKED ENDPOINTS

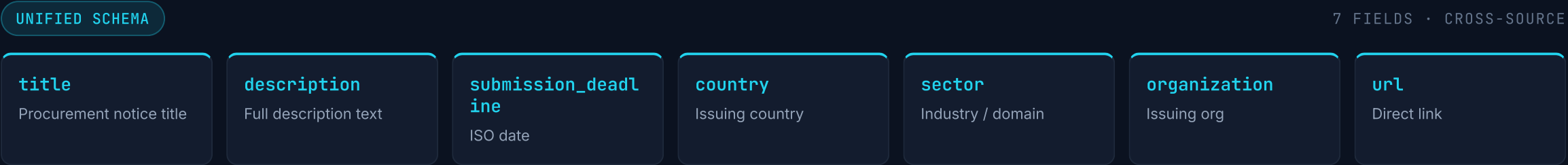
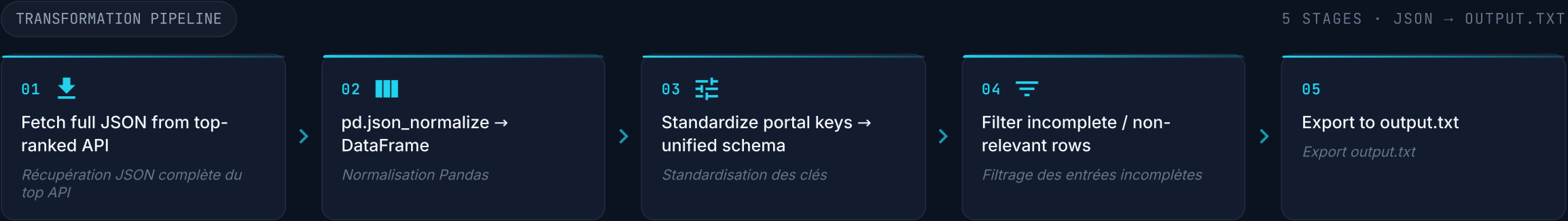
COSINE · TOP 5

/api/v2/procurements/opportunities	SELECTED	0.873
/api/v1/notices/list		0.742
/api/feed/ami		0.681
/api/menu/items		0.234
/api/static/menu		0.119

```
cosine_similarity(query_vec, endpoint_sample_vec) → rank descending
```

Data Cleaning & Schema Normalization

Nettoyage, normalisation & filtrage



Raw portal keys (proc_title, descr, ddv, pays...) mapped to unified schema for cross-source searchability.

Vector Storage & Semantic Search

Stockage vectoriel & recherche sémantique Qdrant

QDRANT COLLECTION SETUP



Auto-init Qdrant collection with Cosine distance metric

Auto-initialisation collection (cosinus)



MD5 hash of description text → idempotent point IDs

Hash MD5 → IDs idempotents



Payload metadata attached: id, title, description

Métadonnées attachées au vecteur

```
collection: procurement_ami
vectors: { size: 384, distance: "Cosine" }
point_id   = md5(description)[:16]
payload    = { id, title, description }
```

SEMANTIC QUERY DEMO

TOP 3 · COSINE

🔍 query: "health" ■

Medical IT infrastructure upgrade

[Country: Tunisia]

0.84

Hospital equipment procurement

[Country: Morocco]

0.79

Public health data platform

[Country: Senegal]

0.71

No keyword match required – pure semantic similarity

Conceptual Data Model (MCD) & Persistence

Modèle conceptuel de données & persistance

 RELATIONAL ENTITIES (MCD)

 USER	 SOURCES	 OPPORTUNITIES
id int PK	id int PK	id int PK
email varchar	organization_name varchar	source_id int FK → Sources
alerts_config json	url text	title varchar
	type enum	document_url text
	scraper_type enum	submission_deadline date
		sector varchar

User accounts & alert subscriptions
Comptes & alertes utilisateur

Configured procurement portals
Portails configurés

Discovered procurement notices
Avis de marchés découverts



 DUAL-LAYER STORAGE

 **SQLite (SQLAlchemy ORM)**
Relational metadata — entities, relationships, deadlines, audit trail
Métadonnées relationnelles



 **Qdrant Vector DB**
384-dim semantic vectors — fast cosine similarity search
Vecteurs sémantiques 384-dim

Phase 1 Results & Deliverables

Résultats & livrables Phase 1

4 / 4 MILESTONES DELIVERED



M1

Live network interception on complex portals

Successfully intercepted XHR/Fetch responses on World Bank Opportunities and other international portals.

Interception réussie sur les portaux complexes (World Bank, etc.)



M2

Automated API ranking by semantic score

Cosine similarity ranking correctly identifies the best data-provider endpoint without hardcoded paths.

Classement automatique des APIs par score sémantique



M3

Standardized data transformation pipeline

Raw JSON → unified 7-field schema → filtered dataset exported as output.txt.

Pipeline de transformation standardisé opérationnel



M4

End-to-end Qdrant embedding & similarity search

Tested insertion of 384-dim vectors and verified semantic retrieval with example queries ("health", "cybersecurity").

Stockage Qdrant & recherche sémantique testés de bout en bout

🚀 All four Phase 1 milestones delivered – system ready for Phase 2 enhancements.

Phase 2 Roadmap & Future Enhancements

Feuille de route Phase 2 & améliorations futures

Q1



Progressive Document Parser

Automated PDF/HTML tender document downloading and structured content extraction.

Téléchargement & extraction automatique des documents d'appel d'offres (PDF/HTML)

STATUS: PLANNED

Q2



NLP Learning

Supervised NLP/LLM module extracting key fields — budget, eligibility, required documents, key dates.

Module NLP/LLM supervisé pour extraction de champs clés

STATUS: PLANNED

Q3



Multilingual Support

Process notices in French, English, and Arabic (FR/EN/AR) with cross-lingual embeddings.

Traitement multilingue FR / EN / AR avec embeddings multilingues

STATUS: PLANNED

Q4



Alerting Interface

FastAPI endpoint expansion, notification system, and stakeholder dashboard UI.

Extension API FastAPI, notifications & tableau de bord

STATUS: PLANNED

AI SCRAPER AGENT • PHASE 2

10 • 10